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A
PROJECT PROPOSAL REPORT
ON
MACHINE LEARNING BASED FLOOD SUSCEPTIBILITY MAPPING OF NARAYANI RIVER BASIN USING REMOTE SENSING AND GIS

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List of Abbreviations

ANN	Artificial Neural Networks
AUC	Area Under the Curve
FSM	Flood Susceptibility Mapping
GEE	Google Earth Engine
GIS	Geographic Information System
LULC	Land Use Land Cover
ML	Machine Learning
NDVI	Normalized Difference Vegetation Index
NRB	Narayani River Basin
RF	Random Forest
ROC	Receiver Operating Characteristic
SPI	Stream Power Index
XGBoost	Extreme Gradient Boosting

1. Introduction

1.1 Background

Floods are among the most frequent and destructive natural hazards worldwide, causing significant losses to human lives, infrastructure, agriculture, and economic activities, particularly in developing countries (1). The increasing impacts of climate change, rapid urbanization, land-use change, and population growth have intensified flood risks in many regions, highlighting the need for effective flood management and mitigation strategies (2). Globally, floods account for nearly 44% of all natural disasters, with Asia experiencing the highest flood occurrence and associated impacts due to rapid population growth, climate variability, and increasing urbanization (2). Consequently, accurate identification of flood-prone areas has become a critical component of disaster risk reduction, sustainable land-use planning, and watershed management (3)(1).

Flood susceptibility mapping is a proactive approach that identifies areas with varying likelihoods of flood occurrence based on the spatial relationship between historical flood events and environmental conditioning factors. Advances in Geographic Information Systems (GIS) and Remote Sensing (RS) technologies have substantially improved the ability to analyze spatial factors influencing flood occurrence and to generate accurate susceptibility assessments over large areas (4). Early flood susceptibility studies primarily relied on statistical approaches and GIS-based multi-criteria decision-making methods to evaluate flood-prone regions and determine the influence of environmental factors on flood occurrence (5). These approaches provided valuable insights into flood risk assessment; however, their predictive capability was often constrained by subjective weighting procedures and difficulties in representing complex environmental interactions (6). To address these limitations, researchers increasingly adopted machine learning techniques that can effectively model nonlinear relationships among flood conditioning factors and flood events (4). Machine learning algorithms have demonstrated strong potential for flood susceptibility mapping due to their ability to model complex nonlinear relationships between flood occurrences and environmental conditioning factors, resulting in improved prediction accuracy and reliability (7).

Nepal, situated in the central Himalayan region of South Asia, is highly susceptible to flood hazards because of its rugged topography, monsoon-dominated climate, fragile geology, and extensive river network. The country contains more than 6,000 rivers and rivulets with a combined length exceeding 45,000 km, many of which experience recurrent flooding during the monsoon season (June–September). According to recent studies, floods remain one of the most significant natural disasters in Nepal, causing substantial human casualties, infrastructure damage, agricultural losses, and economic disruption every year (8)(6). Recent studies have demonstrated the applicability of GIS, RS, and machine learning techniques for flood susceptibility assessment in various regions of Nepal, including the Karnali River Basin and Kathmandu Metropolitan City (8)(6). Although these studies have contributed significantly to flood risk assessment, comprehensive investigations employing multiple machine learning algorithms within major river basins remain limited (8).

The Narayani River Basin is one of the largest and most socioeconomically important river systems in Nepal. The basin supports agriculture, hydropower development, transportation, tourism, and diverse ecosystems. Nevertheless, recurrent flooding poses significant threats to local communities, infrastructure, and economic activities throughout the basin. Owing to its diverse physiographic conditions, intense monsoonal precipitation, and complex hydrological characteristics, the basin represents an ideal region for flood susceptibility assessment.

Therefore, this study aims to develop flood susceptibility maps for the Narayani River Basin using Remote Sensing, GIS, and machine learning approaches. Flood-conditioning factors including elevation, slope, aspect, curvature, rainfall, soil type, distance to river, NDVI, and Stream Power Index (SPI) will be utilized to model flood susceptibility. Three machine learning algorithms—Random Forest (RF), Artificial Neural Network (ANN), and Extreme Gradient Boosting (XGBoost)—will be trained and evaluated using historical flood inventory data. Model performance will be assessed using AUROC, Accuracy, Kappa Coefficient, and F1-Score. The resulting flood susceptibility maps are expected to provide valuable information for flood risk reduction, land-use planning, watershed management, and disaster preparedness within the Narayani River Basin.

1.2 Problem Statement

Flooding is a recurring natural hazard in the Narayani River Basin, causing significant damage to human lives, infrastructure, and agricultural activities, particularly during the monsoon season.The absence of reliable flood susceptibility maps limits evidence-based decision-making for flood mitigation, land-use planning, and disaster preparedness in the Narayani River Basin.

Although advances in Remote Sensing, GIS, and machine learning have improved flood susceptibility assessment worldwide, their application in the Narayani River Basin has not been adequately explored. In particular, comparative evaluation of

machine learning models such as Random Forest, Artificial Neural Network, and XGBoost for flood susceptibility mapping is lacking.
Therefore, there is a need to develop and assess machine learning-based flood susceptibility models using geospatial and remote sensing data to identify vulnerable areas and support informed decision-making for flood risk reduction and sustainable land-use planning in the Narayani River Basin.

1.3 Objectives

1.3.1 General Objective

The general objective of this study is to develop and evaluate machine learning based flood susceptibility mapping framework using remote sensing data and GIS techniques for the Narayani River Basin in Nepal.

1.3.2 Specific Objectives

- To identify, acquire, and process multi-source remote sensing and geospatial data for the preparation of flood conditioning factor layers including Elevation, Curvature, Aspect, Slope, SPI, NDVI, Rainfall, Soil type, and Distance to River (DTR).
- To prepare a flood inventory map from historical flood occurrence data.
- To develop and compare RF, ANN, and XGBoost models for flood susceptibility mapping.
- To produce and validate flood susceptibility maps for the study area classified into five susceptibility zones (very low, low, moderate, high, very high).

2. Literature Review

Floods are among the most destructive and frequently occurring natural hazards worldwide, causing severe impacts on human life, infrastructure, agriculture, and economic development, and their increasing frequency has been strongly linked with climate change, rapid urbanization, and land-use alterations (2). Flood Susceptibility Mapping (FSM) has therefore become an essential component of disaster risk reduction, as it helps to identify and delineate areas that are more likely to experience flooding in the future, thereby supporting effective planning and mitigation strategies (4).

Recent developments in Remote Sensing (RS) and Geographic Information Systems (GIS) have significantly improved flood susceptibility studies by enabling the integration and analysis of large-scale spatial and temporal datasets related to terrain, hydrology, soil, vegetation, and land use (4). These technologies allow researchers to derive critical flood-conditioning variables such as elevation, slope, drainage density, rainfall distribution, land use/land cover (LULC), soil type, vegetation indices, and proximity to river networks, which are essential for understanding flood behavior in a watershed (8). The combination of RS and GIS has therefore become a standard foundation for modern flood hazard analysis due to its efficiency, accuracy, and spatial coverage capabilities (1).

In recent years, machine learning (ML) techniques have emerged as powerful tools for flood susceptibility mapping due to their ability to model complex nonlinear relationships among flood-conditioning factors and historical flood occurrences (2). Unlike traditional statistical approaches, ML models do not rely heavily on predefined assumptions, making them more flexible and suitable for heterogeneous environmental conditions found in river basins (4). Studies have shown that ML-based models generally provide higher predictive accuracy and improved generalization performance when compared to conventional methods (7).

The integration of machine learning with RS and GIS has further enhanced flood susceptibility assessment by enabling data-driven decision-making and improved spatial prediction of flood-prone zones (1). This integrated framework allows for the processing of multi-source geospatial data and the extraction of hidden patterns that contribute to flood generation processes (8). As a result, ML-based GIS models are increasingly being adopted for regional and basin-scale flood hazard studies due to their robustness and adaptability (4).

Ensemble and hybrid machine learning approaches have gained particular attention in recent research because they combine the strengths of multiple algorithms to improve prediction accuracy and reduce model uncertainty (3). Such approaches have been found to outperform individual classifiers by minimizing bias and variance errors, thereby producing more stable and reliable flood susceptibility maps (1). This demonstrates that model integration plays a crucial role in enhancing predictive performance in complex hydrological environments (3).

Despite the success of machine learning approaches, traditional statistical and multi-criteria decision-making methods such as Frequency Ratio and MCDA are still widely used in flood susceptibility studies due to their simplicity and ease of interpretation (5). However, these methods often involve subjective weighting of factors and may not fully capture nonlinear relationships among variables, which can limit their predictive capability in complex terrain conditions (6). Comparative studies indicate that the performance of different models varies depending on study area characteristics, input data quality, and selected conditioning factors, highlighting the importance of model evaluation and selection (7).

Several studies conducted in Nepal and other regions have demonstrated the effectiveness of machine learning for flood susceptibility mapping in river basins with complex topography and monsoon-driven hydrology (8). In particular, research in Himalayan river systems has shown that topographic variables such as elevation, slope, and distance from rivers play a dominant role in flood occurrence, followed by hydrological and land-use factors (8). These findings highlight the importance of selecting relevant conditioning factors to improve model accuracy and reliability (2).

Although flood susceptibility studies have been conducted in various parts of Nepal, including urban and river basin environments, there is still limited research focusing specifically on the Narayani River Basin using advanced machine learning approaches integrated with RS and GIS techniques. Given the basin's high flood vulnerability due to monsoon rainfall, steep topography, and expanding settlements along floodplains, a detailed and data-driven susceptibility assessment is required (4). Therefore, this study aims to develop a machine-learning-based flood susceptibility map of the Narayani River Basin using Remote Sensing and GIS, with the objective of identifying flood-prone zones and supporting sustainable land-use planning and disaster risk management strategies.

3. Methodology

This study aims to assess and map flood susceptibility in the Narayani River Basin using machine learning techniques integrated with Remote Sensing and GIS. The methodology involves the collection of multi-source geospatial datasets, preparation of a flood inventory map, generation of flood conditioning factor layers, data preprocessing and multicollinearity analysis, development of individual machine learning models (Random Forest, Artificial Neural Network, and XGBoost). The generated flood susceptibility maps will be validated using statistical performance measures such as AUROC, F1-score, and Cohen's Kappa coefficient. Finally, the performance of individual models will be compared to identify the most effective approach for flood susceptibility assessment in the study area.

3.1 Data Acquisition

The flood susceptibility mapping of the Narayani River Basin will be conducted using a combination of remote sensing, GIS, and secondary datasets obtained from reliable sources. A 30 m resolution Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) will be used to derive topographic factors including elevation, slope, aspect, curvature, distance to river (DTR), and Stream Power Index (SPI). Vegetation information will be represented through the Normalized Difference Vegetation Index (NDVI), which will be calculated from Sentinel-2 satellite imagery with a spatial resolution of 10 m. Rainfall data, one of the most important hydrological factors influencing flood occurrence, will be collected from the Department of Hydrology and Meteorology (DHM), Nepal. Soil data will be obtained from the National Soil Science Research Center (NSSRC), Nepal, providing information on soil characteristics that affect infiltration and runoff processes.Historical flood inventory data covering the most recent 10-year period (2016–2025) will be acquired from the National Disaster Risk Reduction and Management Authority (NDRRMA), Nepal. In addition, complementary flood records will be collected from the Department of Hydrology and Meteorology (DHM), Nepal, the DesInventar disaster database, and global datasets such as the Dartmouth Flood Observatory (DFO) and Copernicus Emergency Management Service (EMS) to improve data completeness and validation. These flood occurrence records will serve as the dependent variable for machine learning model training and validation. These datasets will be preprocessed, standardized, and integrated within a GIS environment to develop flood susceptibility models using Random Forest (RF), Extreme Gradient Boosting (XGBoost) and Artificial Neural Network (ANN) algorithms.

Table 3.1: Data Source to be Used

S.N.	Data	Source	Resolution
1	Digital Elevation Model (DEM)	SRTM	30 m
2	NDVI	ESRI Sentinel-2	10 m
3	Rainfall	DHM, Nepal	Station Data
4	Soil	National Soil Science Research Center	100 m
5	Distance to River	Generated from river network	30 m
6	SPI, Aspect, Slope, Curvature	Generated from DEM	30 m
7	Flood Inventory	NDRRMA	Point Data

3.2 Software and Tools

The study will utilize a combination of GIS, remote sensing, and machine learning tools for data processing, analysis, and flood susceptibility modeling. ArcGIS and QGIS will be used for spatial data preprocessing, thematic layer generation, and map preparation. Google Earth Engine will be employed for processing Sentinel-2 imagery and deriving NDVI. Machine learning models, including Random Forest (RF), XGBoost, and Artificial Neural Network (ANN), will be developed using Python and its associated libraries. Model validation and statistical analyses, including Pearson Correlation, VIF, AUROC, F1-score, Kappa Coefficient, and Accuracy assessment, will also be performed within the Python environment. This integrated framework will facilitate efficient flood susceptibility assessment in the Narayani River Basin.

3.3 Study Area

The Narayani River Basin, also known as the Gandaki River Basin, is one of the major river basins of Nepal. Located in central Nepal, the basin extends approximately between 82.88°E to 85.70°E longitude and 27.36°N to 29.33°N latitude. The basin covers an area of approximately 31,500 km^2 and drains a large portion of central Nepal before flowing southward into India. The basin is formed by the confluence of several major river systems, including the Kali Gandaki, Trishuli, Marsyangdi, Seti, and Budhi Gandaki rivers. It encompasses parts of several districts, including Mustang, Manang, Myagdi, Baglung, Kaski, Parbat, Syangja, Tanahun, Lamjung, Gorkha, Rasuwa, Nuwakot, Dhading, Palpa, Chitwan, Nawalparasi, and Makawanpur. The drainage network of the basin is highly developed, consisting of numerous tributaries that contribute to the Narayani River system. The basin exhibits significant physiographic diversity, ranging from the High Himalayan region in the north to the low-lying Terai plains in the south. Elevation varies from less than 100 m above mean sea level in the southern floodplains to over 8,000 m in the northern Himalayan region. This substantial variation in topography strongly influences hydrological processes such as runoff generation, infiltration, sediment transport, and river discharge. The climate of the basin is predominantly monsoonal, with nearly 80% of annual precipitation occurring during the monsoon season from June to September. Intense rainfall events during this period frequently result in high river discharge, flash floods, riverbank erosion, and inundation in downstream areas. Districts such as Chitwan and Nawalparasi are particularly vulnerable to flood hazards due to low terrain gradients, river channel migration, and sediment deposition along floodplains. In addition to natural factors, anthropogenic activities such as deforestation, agricultural expansion, infrastructure development, and settlement growth within flood-prone areas have increased flood vulnerability in recent years. Due to its diverse topography, extensive drainage network, varied climatic conditions, and recurring flood events, the Narayani River Basin provides an ideal setting for flood susceptibility assessment. Therefore, the integration of Remote Sensing, Geographic Information Systems (GIS), and Machine Learning techniques can effectively identify flood-prone areas and support disaster risk reduction, watershed management, and sustainable land-use planning within the basin.

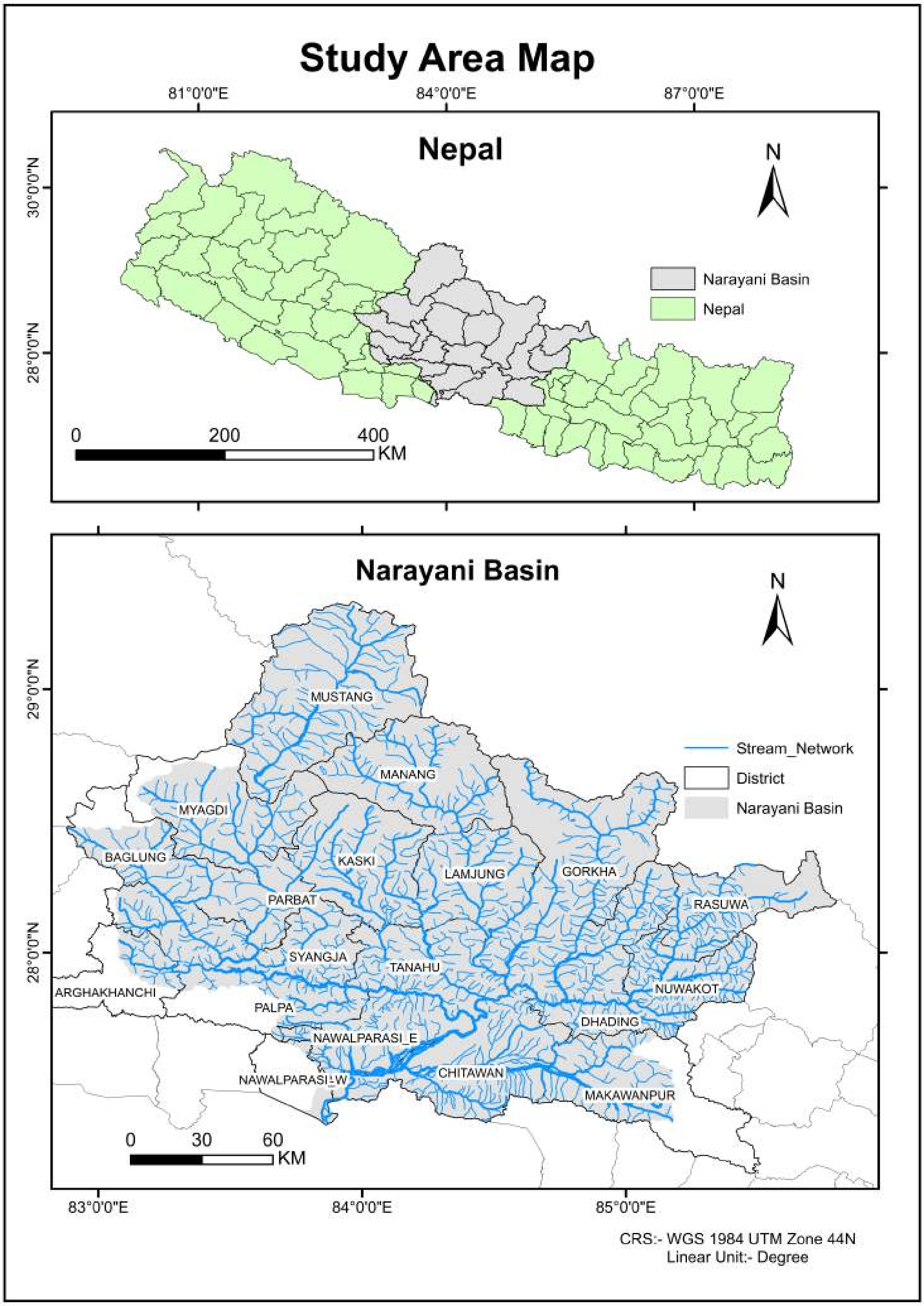

Figure 3.1: Study Area

3.4 Methods and Workflow

The methodology adopted in this study integrates Remote Sensing (RS), Geographic Information System (GIS), and Machine Learning (ML) techniques to assess flood susceptibility in the Narayani River Basin. Initially, spatial and non-spatial datasets including SRTM DEM, Sentinel-2 imagery, rainfall data, soil data, and flood inventory records will be collected from respective sources. The acquired datasets will undergo preprocessing, including projection, clipping, resampling, and rasterization to ensure uniform spatial resolution and coordinate systems. Subsequently, flood conditioning factors, namely elevation, slope, aspect, curvature, distance to river (DTR), Stream Power Index (SPI), NDVI, rainfall, and soil, will be prepared and standardized within a GIS environment. Elevation, slope, aspect, curvature, DTR, and SPI will be derived from the SRTM DEM, while NDVI will be generated from Sentinel-2 imagery. Rainfall and soil datasets will be obtained from DHM and NSSRC, respectively. Flood inventory data acquired from NDRRMA will be used as the dependent variable for model training and validation.

Before model development, multicollinearity among the conditioning factors will be evaluated using Pearson Correlation Analysis and Variance Inflation Factor (VIF). Variables exhibiting strong multicollinearity will be removed to improve model performance. Following feature selection, the flood inventory dataset will be divided into training (70%) and testing (30%) subsets.

Three machine learning algorithms, namely Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Artificial Neural Network (ANN), will be employed to develop flood susceptibility models. The trained models will generate susceptibility indices, which will subsequently be classified into five categories: Very Low, Low, Moderate, High, and Very High susceptibility zones.

Finally, model performance will be evaluated and compared using Accuracy, F1-Score, Kappa Coefficient, and Area Under the Receiver Operating Characteristic Curve (AUROC). The best-performing model will be selected based on validation results and used to produce the final flood susceptibility map of the Narayani River Basin.

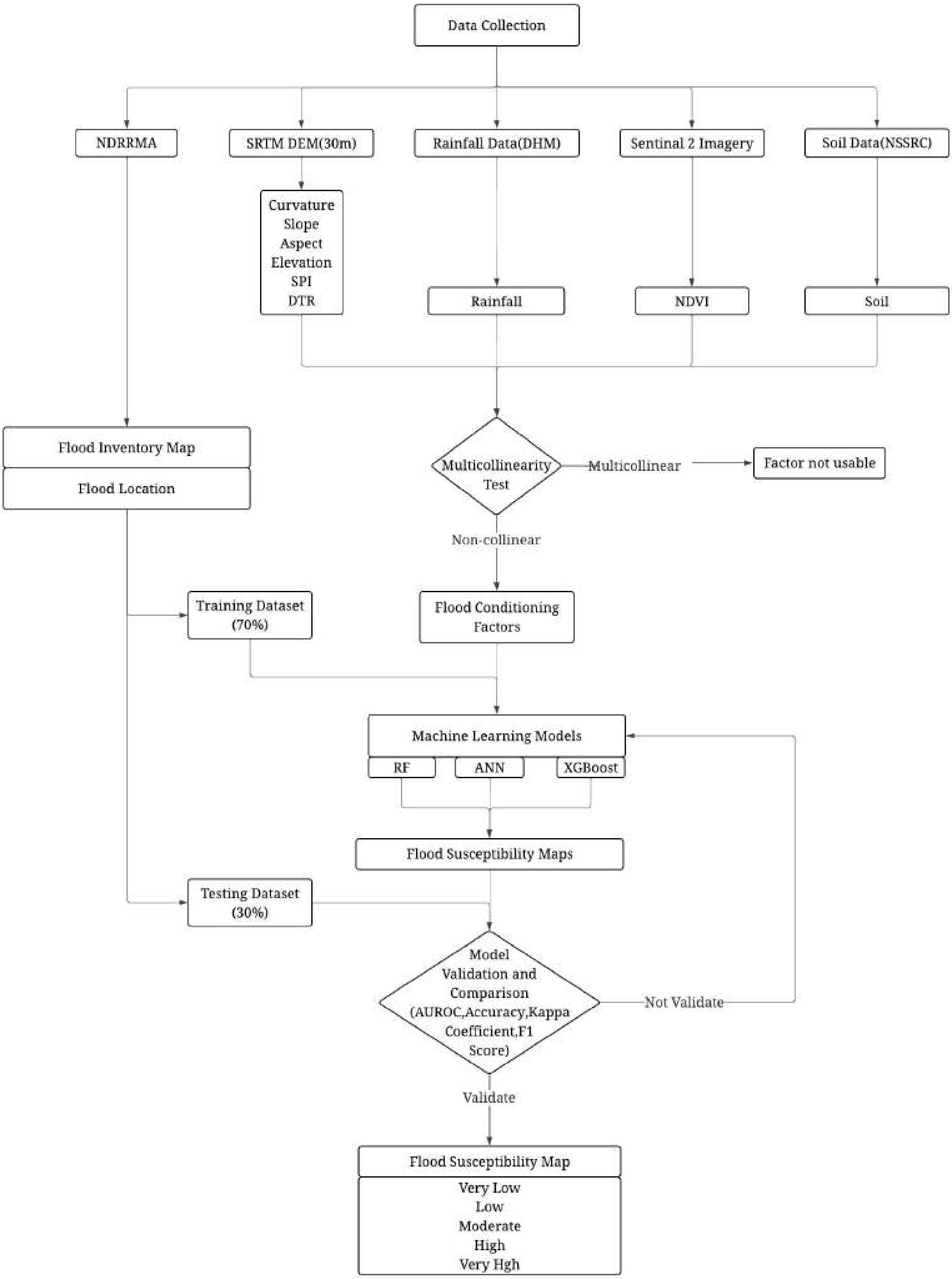

Figure 3.2: Methodology Flowchart

4. Expected Outcomes

1. Preparation of a Flood Susceptibility Map (FSM) of the Narayani River Basin using Remote Sensing, GIS, and Machine Learning techniques.

2. Determination of the most significant flood conditioning factors such as elevation, slope, rainfall, distance to river, and NDVI.

3. Development and comparison of machine learning models for flood susceptibility assessment.

4. Identification and classification of flood-prone areas into Very Low, Low, Moderate, High, and Very High susceptibility zones.

5. Work Schedule

Figure 5.1: Work Schedule

Figure 5.1 illustrates the proposed work schedule for the research project spanning from June to April. The study commences with proposal preparation and submission in June. Concurrently, the literature review and documentation process continues throughout the entire study period to ensure continuous refinement and support of the research work.

Data collection is scheduled from July to September, followed by data processing during October and November. The analysis and validation phase will be carried out in December and January, during which the developed machine learning models will be evaluated and results will be assessed. Based on the findings, necessary improvements and refinements will be implemented from February to March. Finally, the project will conclude with the preparation and submission of the final report between March and April.

Overall, the schedule provides a systematic and well-structured framework for completing all research activities within the allocated timeframe.

6. Estimated Budget

The estimated budget for the proposed flood susceptibility mapping study is NPR 20,000. The budget covers expenses related to data collection, field transportation, model validation, printing and documentation, and miscellaneous costs. A major portion is allocated for transportation to support field verification and data acquisition activities. This budget is expected to adequately support the successful implementation and completion of the research project.

Table 6.1: Budget Estimation Table

SN	**Particulars**	**Amount (NPR)**
1	Data Collection	2000
2	Transportation	10000
3	Model Validation	2000
4	Printing and Documentation	4000
5	Miscellaneous	2000
	Total	**20000**

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